INDIAN INSTITUTE OF ASTROPHYSICS

Night Sky over Karnataka

A beginner's guide to the stars you can see from your village

IIA Outreach Team

2023 · English · CC BY-NC 4.0

Arivu Kendra sample edition · Government of Karnataka

Contents

1. Why the sky is dark here 3
2. Finding your way: north, east and the pole star 9
3. The planets this year 16
4. The Milky Way 22
5. Nakshatras and the Kannada almanac 29
6. Twelve sky maps 35
7. Keeping an observation diary 42

This is a generated sample edition created for the Arivu Kendra public preview. The text on these pages is placeholder prose so that reading tools, previews and downloads can be demonstrated end to end. The real edition will be supplied by the content provider.

CHAPTER 1

Why the sky is dark here

Rural Karnataka has some of the darkest skies in southern India. This guide, written for Arivu Kendra readers, explains what you can see with your eyes alone: the planets, the brightest stars, the Milky Way in winter and the constellations the Kannada almanac has named for centuries. Each chapter ends with a sky map for that month.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred

villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have

used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At

sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside

will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

CHAPTER 2

Finding your way: north, east and the pole star

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them

apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room

is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned

before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of

your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the

whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

CHAPTER 3

The planets this year

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your

phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays

together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright.

This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred

villages is data, and data is how we learn things nobody has learned before.

CHAPTER 4

The Milky Way

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in

Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between

the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina

rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the

telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres

away, can still burn your skin.

CHAPTER 5

Nakshatras and the Kannada almanac

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time,

where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from

the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a

small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty

minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

CHAPTER 6

Twelve sky maps

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this

one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it.

Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single

observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in

Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between

the tropics, it shrinks to nothing at all.

CHAPTER 7

Keeping an observation diary

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your

phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays

together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright.

This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred

villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have

used this trick for thousands of years, and it still works.